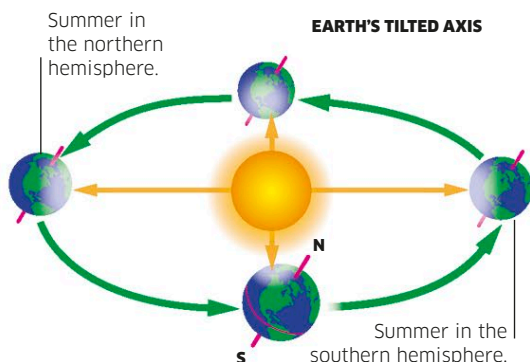


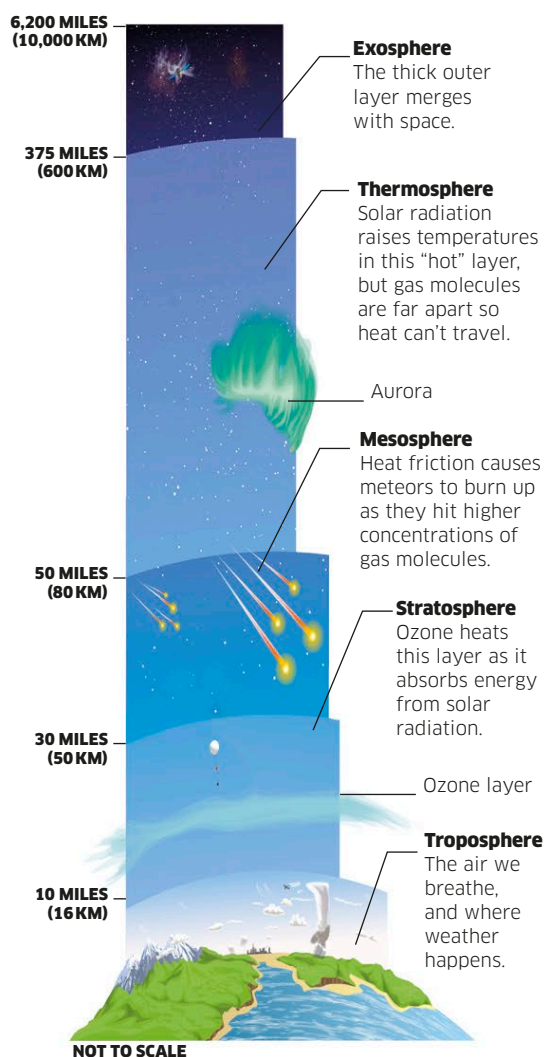
The seasons

As Earth orbits around the sun, it also rotates around its axis—an imaginary north-south line. This axis is tilted by 23.4° compared to Earth's orbit, so that one part of the planet is always closer to or farther away from the sun, resulting in the seasons.



Atmosphere

Earth's atmosphere is made up of a mix of gases—78 percent nitrogen, 21 percent oxygen, and a small amount of others, such as carbon dioxide and argon. These gases trap heat on the planet and let us breathe. The atmosphere has five distinct layers.



Earth and Moon

Our home, Earth, is about 4.5 billion years old. With a diameter of just over 7,500 miles (12,000 km), it orbits the sun every 365.3 days and spins on its axis once every 23.9 hours.

Of all the planets in the universe, ours is the only place life is known to exist. Earth is one of the solar system's four rocky planets, and the third from the sun. Its atmosphere, surface water, and magnetic field—which protects us from solar radiation—make Earth the perfect place to live.

Inside Earth

Earth is made up of rocky layers. The outer crust floats on a rocky shell called the mantle. Beneath this is the hot, liquid outer core and solid, inner core.

Outer core

The liquid outer layer of the Earth's core is hot. Made of liquid iron and nickel, it is 1,400 miles (2,300 km) thick.

Oceanic crust

The solid outer layer of rocks is the crust. Under the oceans, it is only about 6 miles (10 km) thick, but it is denser than the continental crust.

Continental crust

The continental crust is the land on which we stand. It is much thicker than the oceanic crust—up to 45 miles (70 km) thick—but is less dense.

Sun

The sun's diameter is 109 times Earth's.

